Updating Rows with UPDATE

Lecture Notes · Game DB Junior

In this lesson

- How to change data in a row with **UPDATE** and **SET**
- How **WHERE** chooses which rows to update
- Why the WHERE statement must be **specific enough**

1. Updating and deleting

Now that we understand WHERE, we look at two more things we often do on tables: **updating** records and **deleting** records. In this lesson we start with updating: how to change properties on a row in a table.

2. Example: Tokyo's population

Tokyo has a population of about 38 million. Maybe in a year there will be one million more people. So we want to update it to about **39 million**.

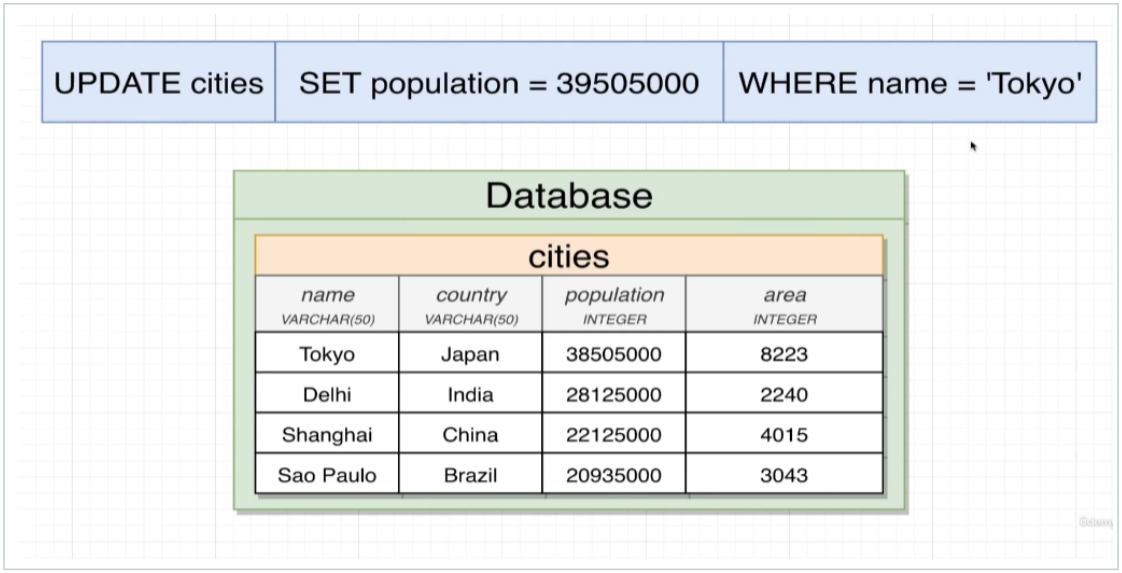


Figure 1 — UPDATE cities SET population = 39505000 WHERE name = 'Tokyo'

```
UPDATE cities
SET population = 39505000
WHERE name = 'Tokyo';
```

This query looks very different from a SELECT statement. There is **no SELECT**, because we are not fetching any data. We only want to update some data.

Part	What it does
UPDATE	Keyword: we want to update data
cities	The name of the table we want to change

Part	What it does
SET population = 39505000	SET is a new keyword. It says which property we want to change on a record, and the new value
WHERE name = 'Tokyo'	Says exactly which records we want to update

We only have one row with the name Tokyo. So this finds that one row and updates its population to 39 million.

3. Checking the change

- When we run the UPDATE, we only get a message that the update was **successful**.
- To see what changed, we must run a query that fetches all the rows: **SELECT * FROM cities;**
- Now we can see that Tokyo has a population of 39 million.

4. Be careful with WHERE

The WHERE statement can be a bit **tricky**. If you want to update one specific record, your WHERE statement must be **specific enough** to update only that one city.

Some cities in the world are named after other cities. For example, there are cities in the **United States** called **Shanghai**, the same name as Shanghai in China. If our table grows, we may have two rows with the name Shanghai.

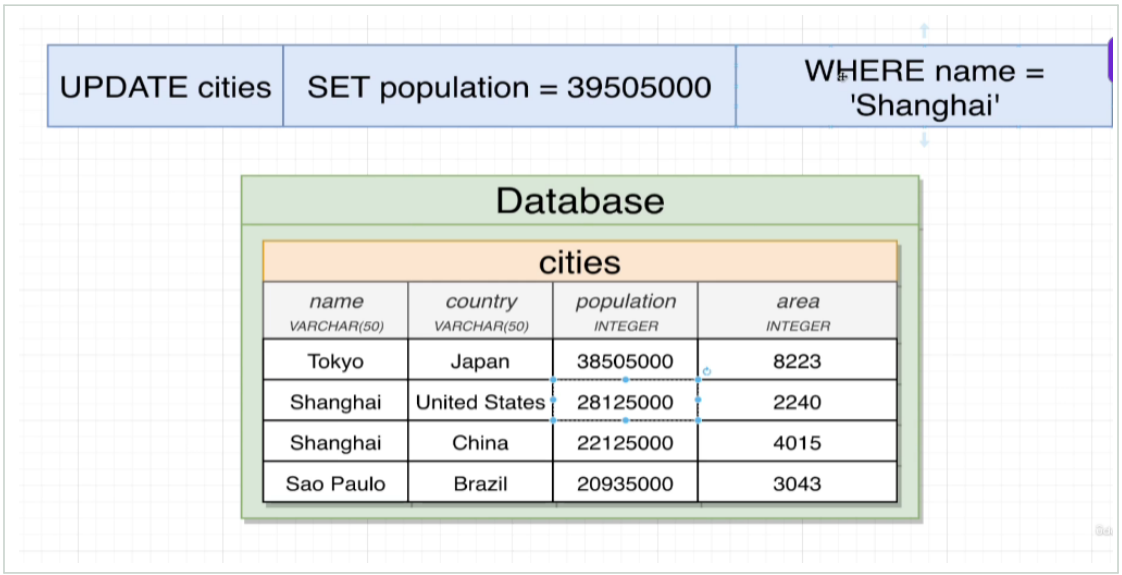


Figure 2 — Two rows named Shanghai: one in the United States, one in China

```
UPDATE cities
SET population = 39505000
WHERE name = 'Shanghai';
```

- We want to update only Shanghai in China.
- But this query updates **both** rows: Shanghai in China **and** Shanghai in the United States.

- So be very careful when you write WHERE statements.

> **Don't worry:** later in this course we will learn a foolproof way to make sure an update always changes exactly what we want.

UPDATE is simple. It is rare to write really complicated UPDATE statements. Most of them are simple, like the one above.

5. Running the code (my practice)

Step 1: update Tokyo's population to 39505000.

```sql
-- Change the population of Tokyo to 39505000
UPDATE cities
SET population = 39505000
WHERE name = 'Tokyo';  -- only update the row where the name is Tokyo
```

Figure 3 — UPDATE with comments

The Messages tab shows **UPDATE 1** and "Query returned successfully". The **1** is the number of rows that were updated.

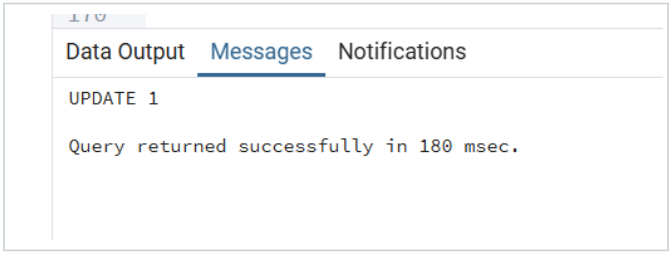
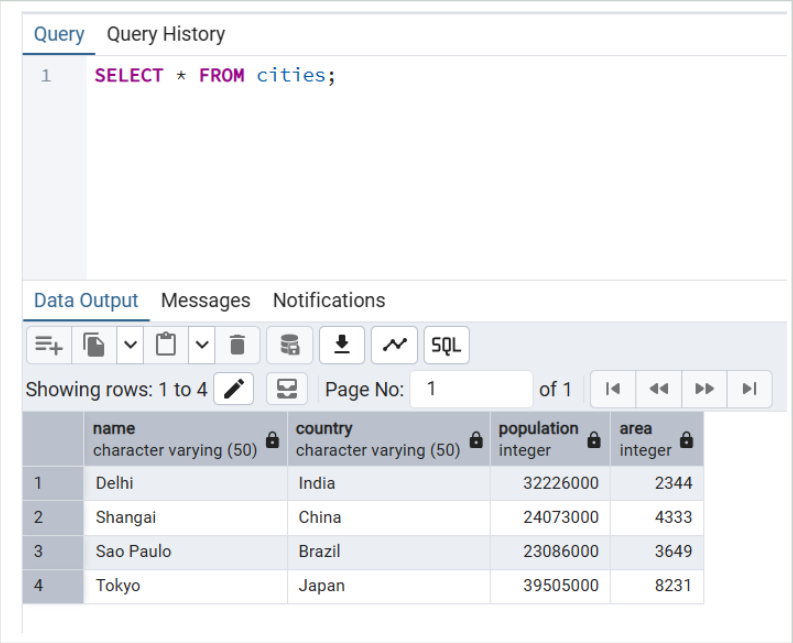


Figure 4 — UPDATE 1

Step 2: check the change with SELECT * FROM cities. Tokyo now has a population of **39505000**.



	name character varying (50)	country character varying (50)	population integer	area integer
1	Delhi	India	32226000	2344
2	Shangai	China	24073000	4333
3	Sao Paulo	Brazil	23086000	3649
4	Tokyo	Japan	39505000	8231

Figure 5 — Tokyo's new population

6. Summary

Idea	Remember
UPDATE table	Choose the table we want to change
SET column = value	Choose the property to change and the new value
WHERE ...	Choose exactly which rows to update
Check the result	UPDATE only says it was successful. Use SELECT to see the change
Be specific	If WHERE matches more rows than you want, all of them are updated

Next lesson: how to delete rows with DELETE.

Sources
Lecture transcript and lecture images (Figures 1–2)
My own practice screenshots (Figures 3–5)